GLANCE ML INTERNSHIP ASSIGNMENT

Multimodal Fashion & Context Retrieval

Natural-language fashion image search across garments, style, and scene context

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GitHub repository	[GITHUB REPOSITORY LINK]
Demo video	[DEMO VIDEO LINK]

1 Problem & Design Goal

The task is to build a search engine over a fashion image corpus that answers free-text queries combining *what* is worn (garments, colors), the *vibe* (style), and *where* it is worn (scene context) — for example *“a bright yellow raincoat”*, *“a red tie and a white shirt in a formal setting”*, or *“someone wearing a blue shirt sitting on a park bench”*.

The central ML problem: global CLIP is a strong zero-shot semantic baseline — our own measured vanilla baseline reaches Precision@5 of 0.87 on this corpus — but it compresses the whole image and the whole query into single vectors. Fine-grained garment evidence (a small tie occupying 2% of the frame) can be diluted, and compositional constraints can bind loosely: a query mentioning “red” and “tie” can be satisfied, in embedding space, by a red outfit with no tie.

Design question: can fashion-specific and region-level evidence improve top-k relevance on fine-grained and compositional queries while retaining CLIP’s broad zero-shot semantic capability?

2 Approaches Considered & Trade-offs

A. Vanilla global CLIP retrieval

Encode every image and the full query with CLIP; rank by cosine similarity. **Strengths:** simple, strong zero-shot semantics, trivially scalable, no training. **Weaknesses:** one vector per image and per query; garment-level evidence is diluted; attribute-to-garment binding is implicit and can fail on compositional queries.

B. FashionCLIP-only global retrieval

Replace CLIP with FashionCLIP (a CLIP model adapted to fashion data by its authors; we use it frozen). **Strengths:** a fashion-specialized embedding space with better garment vocabulary and similarity. **Weaknesses:** weaker scene/context specialization than general CLIP; still a single global vector; individual garment clauses are still not explicitly verified.

C. Learned reranker / cross-encoder / fine-tuning

Strengths: potentially stronger relational and compositional reasoning. **Weaknesses:** needs training data and relevance labels that do not exist at this corpus scale; higher retrieval-time cost; disproportionate to a 1,000-image assignment dataset. Kept as future work.

D. Decomposed multi-signal retrieval with region evidence (chosen)

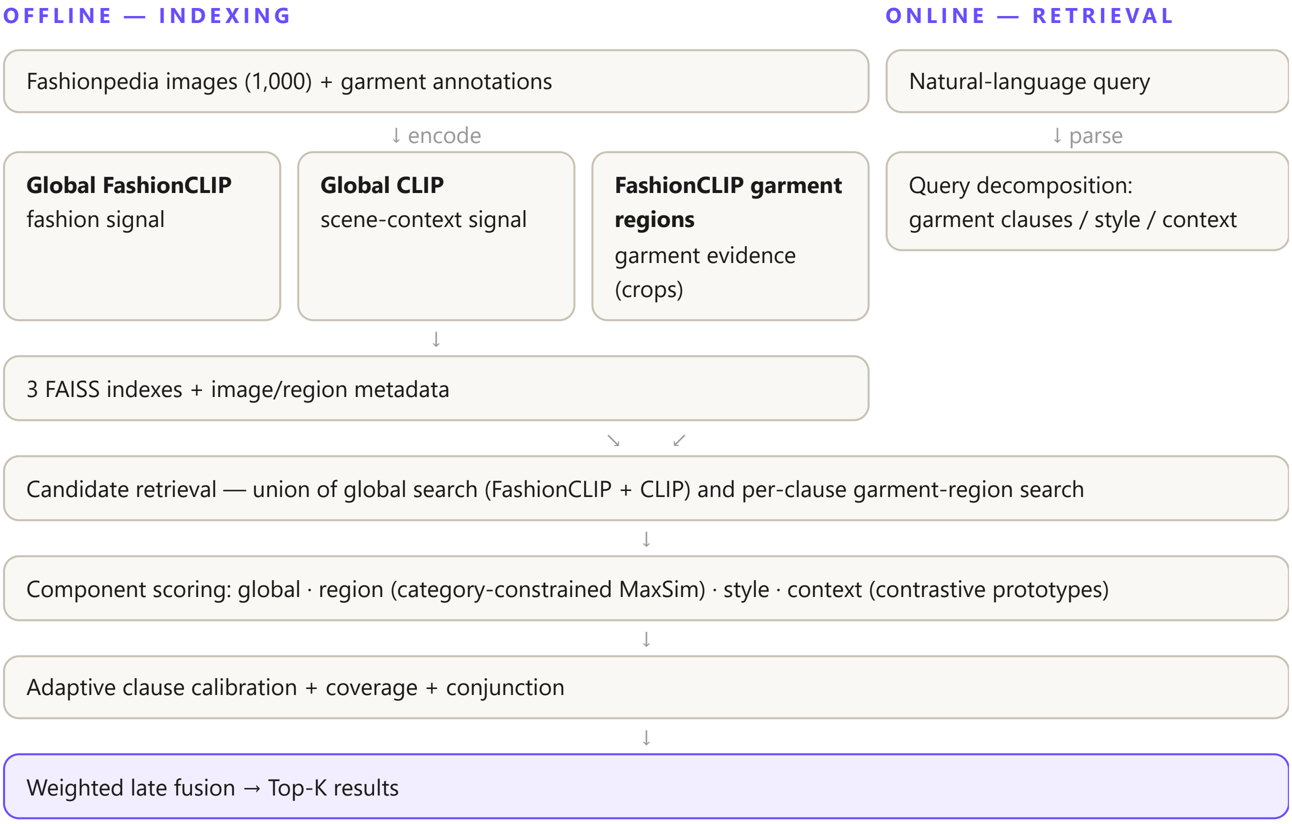
Keep the frozen zero-shot encoders, but use each where it is strongest and add garment-region grounding: FashionCLIP for fashion semantics, general CLIP for scene context, and FashionCLIP embeddings of Fashionpedia-annotated garment crops as explicit per-garment evidence. Signals are combined by query-adaptive late fusion. This preserves zero-shot capability, verifies garment clauses against the right image regions, stays interpretable (each result carries per-component scores), and everything remains FAISS-indexable.

3 Chosen Architecture

Offline indexing

For each of the 1,000 Fashionpedia images: encode the full image with FashionCLIP and with general CLIP (open_clip ViT-B/32); crop every annotated garment bounding box (~4,000 regions across 27 categories) and encode each crop with FashionCLIP. All vectors are L2-normalized and stored in flat inner-product FAISS indexes, with metadata CSVs mapping every vector and region back to its source image and Fashionpedia category.

Three evidence sources exist because they answer different questions: the FashionCLIP global signal captures overall fashion appearance; the general CLIP signal captures broad scene/context semantics; the region FashionCLIP signal provides garment-specific evidence that survives even when a garment is small in the frame.



Query decomposition

A lightweight rule-based parser splits the query into garment clauses, style terms, and context terms. *"A red tie and a white shirt in a formal setting"* becomes the clauses *red tie* (Fashionpedia category *tie*) and *white shirt* (categories *shirt/blouse, top*), plus a *formal* style signal. No learning is involved in this step.

Region candidate retrieval

Each recognized clause is encoded with FashionCLIP and searches the garment-region index directly. A small matching garment can therefore pull its parent image into the candidate pool even when the whole-image embedding ranks poorly for the query — the candidate pool is the union of global FashionCLIP, global CLIP, and per-clause region hits.

Category-constrained clause-to-region matching

When scoring a candidate image, each clause is compared only against that image's regions of compatible Fashionpedia categories: a *tie* clause is scored against tie crops, not arbitrary crops. The clause score is the maximum similarity over compatible regions (MaxSim).

Coverage, conjunction, and adaptive calibration

Multi-garment queries are rewarded for satisfying *all* requested constraints rather than one strong match: a coverage score counts matched clauses and a conjunction bonus fires only when every clause matches. Whether a clause "matches" is decided adaptively — against the percentile of that clause's own score distribution across the current candidate pool (with an absolute floor) — rather than one universal cosine threshold, because rare garment categories produce systematically different score ranges than common ones.

Style, context, and late fusion

Style terms are scored with FashionCLIP against the whole image. Context terms are scored with general CLIP; for known places (office, park, city, street, home) a contrastive prototype bank is used — mean similarity to positive prompts ("professional office interior") minus a weighted maximum over negative prompts ("outdoor street", "fashion runway") — which gives a sharper scene signal than a single embedding. Finally, only the components the query actually contains are activated, min-max normalized over the candidate pool, and combined:

$$score(i) = \Sigma_{c \in active}\, w_c \cdot \tilde{s}_c(i) \;/\; \Sigma_{c \in active}\, w_c$$

where $\tilde{s}_c$ are normalized component scores (global, region, coverage, conjunction, style, context) and w_c are fixed hand-set weights, renormalized over active components. A garment-free query like "casual outfit in an urban street" therefore reduces gracefully toward global + style + context retrieval.

4 Quantitative Evaluation: Final System vs Vanilla CLIP

Both systems were evaluated on the same 15 queries (the five mandatory assignment queries plus ten additional garment / compositional / style / context queries), top 5 results per query, with human relevance judgments on a 0 / 1 / 2 scale (irrelevant / partially relevant / highly relevant) under the same rubric. Relevance labels are shared between systems only for identical (query, image) pairs; results retrieved by only one system were labeled independently under the same rubric. Metrics: Precision@5 (relevance ≥ 1), mAP bounded to the retrieved top-5 pool (not corpus-wide AP), and nDCG@5 on graded labels. The vanilla baseline performs its own independent global-CLIP candidate retrieval and ranking; it never sees candidates discovered by the final system.

System	Precision@5	mAP (top-5 pool)	nDCG@5
Vanilla global CLIP (whole-query, whole-image)	0.8667	0.9644	0.9779
Final system (decomposed multi-signal + region evidence)	**0.9333**	0.9631	0.9632

Precision@5 improves from 86.67% to 93.33% (+6.66 percentage points) — the final system places measurably more relevant images in the top five. mAP is essentially unchanged (0.9644 vs 0.9631). nDCG@5 decreases slightly (0.9779 vs 0.9632): where vanilla CLIP does retrieve relevant images, it occasionally orders them slightly better.

One property of pool-bounded nDCG is worth stating because it explains much of this gap: nDCG is normalized against the ideal ordering of the *retrieved* set itself. On “yellow raincoat”, the vanilla baseline retrieves only two relevant images out of five yet scores nDCG = 1.0, because those two happen to rank first. A system can therefore score perfect nDCG while missing most relevant results. In this setup Precision@5 is the more discriminative measure of retrieval quality, and it is where the architectural changes show.

5 The Five Mandatory Assignment Queries

Assignment query	Type	Vanilla P@5	Final P@5
“A person in a bright yellow raincoat”	Attribute	0.4	**0.8**
“Professional business attire inside a modern office”	Contextual	1.0	0.8
“Someone wearing a blue shirt sitting on a park bench”	Complex semantic	0.8	0.8
“Casual weekend outfit for a city walk”	Style inference	1.0	1.0
“A red tie and a white shirt in a formal setting”	Compositional	0.6	**0.8**

Pattern. The strongest gains occur exactly on the fine-grained attribute query (yellow raincoat: 0.4 → 0.8, doubling the relevant results) and the compositional query (red tie + white shirt: 0.6 → 0.8) — the two failure modes the region-grounded architecture directly targets. On the compositional query, the final system's top results include the sparse annotated tie examples available in the corpus, and the conjunction mechanism rewards candidates with evidence for multiple requested garment constraints.

The office regression, honestly. Vanilla CLIP scores 1.0 on the office query; the final system scores 0.8. The corpus (runway- and editorial-heavy Fashionpedia imagery) has limited genuine modern-office coverage, and late fusion cannot recover visual evidence that is sparse in the candidate corpus; the added fashion-side components slightly diluted a query that pure scene semantics already answered well.

The park-bench query. Both systems reach 0.8. Fashion + broad scene context retrieval works, but the exact relation "*sitting on* a bench" is inferred from embedding similarity rather than structurally verified — a stated limitation (Section 7).

Qualitative side-by-side retrieval results for all five queries are shown in the demo video (Section 9); earlier saved result grids predate the final retrieval configuration and are deliberately not reproduced here.

6 Experimentation & Model Selection: Color-Contrast Scoring

Manual inspection of labeled results showed color-binding errors: a visibly red jacket scored almost identically to genuinely blue jackets for the clause "blue jacket", because CLIP-family crop similarity is dominated by garment shape rather than hue. A contrastive color penalty was implemented and tested: each "{color} {garment}" clause score is discounted by how well the same best-matching crop also matches every *other* color for that garment.

The full A/B (weight 0.50 vs disabled), evaluated with the same human-labeling protocol (artifacts under `evaluation/experiments/color_contrast/`), measured a genuine trade-off:

- Compositional nDCG@5 improved 0.912 → 0.945, and a confirmed red-jacket false positive was eliminated from the blue-jacket query.
- Overall Precision@5 decreased 0.933 → 0.907.
- Yellow-attribute queries regressed, concentrated at the yellow/beige/cream visual boundary, which these embeddings cannot separate reliably.

Decision: the feature is disabled in the submitted system (`COLOR_CONTRAST_WEIGHT = 0.0`). It improved one slice at the cost of overall precision — measured model selection rather than intuition. The implementation and its labeled evidence are preserved in the repository for reproducibility, clearly marked experimental.

7 Limitations

- **Corpus coverage.** Fashionpedia provides strong garment annotations, but the 1,000-image corpus has limited modern-office coverage and sparse long-tail garments (two annotated ties in the entire corpus), which bounds what any fusion logic can retrieve.
- **Relations and actions.** "Sitting on a park bench" is inferred through embedding similarity, not structurally verified; there is no pose or spatial-relation model.
- **Closed-vocabulary decomposition.** The parser recognizes a bounded garment/style/context vocabulary; unknown terms fall back to whole-query zero-shot retrieval, losing region grounding but

not failing.

- **Annotation dependence.** Region evidence is bounded by Fashionpedia annotation and category completeness; unannotated garments contribute no region signal.
- **Evaluation scale.** 15 queries, 75 judgments, one relevance labeler; per-category slices are small.
- **Metric scope.** mAP is top-5/pool-bounded, not corpus-wide AP; nDCG is normalized within the retrieved pool (see Section 4).
- **Calibration.** The adaptive clause-match percentile was chosen after inspecting evaluation-set score distributions, not on a held-out calibration split.
- **Fusion.** Component weights are hand-set, not learned.

8 Scalability, Zero-Shot Capability & Future Work

Scaling to ~1 million images

The current flat FAISS indexes give exact nearest-neighbor search and are the right choice at 1,000 images (the assignment explicitly de-prioritizes index engineering). At ~1M images the same retrieval logic carries over with engineering substitutions: replace flat global indexes with IVF or HNSW (optionally PQ-compressed); shard garment-region vectors by Fashionpedia category so a *tie* clause searches only the tie shard after query decomposition; batch and distribute offline encoding; move metadata to persistent storage; and, if needed, adopt an explicit two-stage candidate-retrieval + reranking split. Region reranking is already bounded to the candidate pool rather than the corpus, so query-time cost grows with candidates, not corpus size. The decomposition and fusion logic itself does not depend on the 1,000-image scale.

Zero-shot capability

Both encoders are used frozen, so natural-language zero-shot retrieval is retained end to end. The structured parser strengthens queries whose garments it recognizes, and unknown or unsupported terms still retrieve through whole-query embeddings. Honestly stated: structured region grounding is strongest for vocabulary covered by the parser and the Fashionpedia taxonomy; outside it, the system degrades to (strong) global zero-shot behavior rather than gaining region evidence.

Extending to locations, cities, and places

The late-fusion design extends naturally: add a location-specialist signal (place/landmark-oriented visual-text embeddings or a places-trained classifier) as another component activated only for location-bearing queries, optionally combined with geospatial metadata filtering when images carry location tags. This mirrors how the scene-context component works today — including contrastive positive/negative prototypes per city or place type.

Extending to weather

Weather concepts (rain, snow, sunny, foggy) fit the same pattern: weather-specific text/visual prototypes or a lightweight weather classifier as a query-conditioned fusion component, plus optional metadata joins when timestamp/location exist. Rain in particular pairs well with garment signals (raincoats, umbrellas are already Fashionpedia categories).

Improving precision

- Explicit relation/action detection (sitting, standing, walking, holding) to structurally verify queries like the park-bench prompt.
- Broader garment vocabulary with embedding-based mapping of unseen garment words onto the Fashionpedia taxonomy.
- A learned reranker trained on accumulated human relevance judgments over the top candidate pool, and learned fusion weights replacing hand-set ones.
- Held-out calibration of clause-match thresholds.
- A stronger vision-language reranker applied only to the top ~50 candidates (cost stays bounded).

- More diverse corpus coverage — the office regression and the two-tie sparsity are corpus problems no ranking change can fully fix.

9 Demo Video

A short end-to-end demonstration of the retrieval system is available here: [WATCH DEMO VIDEO]

The demo shows natural-language query input, the parsed garment/style/context interpretation, top-5 ranked retrieval for the assignment queries, and the per-component retrieval evidence behind each result.

10 Codebase & Conclusion

GitHub repository: [GITHUB REPOSITORY LINK]

The repository separates logic from data and indexing from retrieval: `src/indexer/` (encoder wrappers), `src/retriever/` (query decomposition, retrieval, scoring, fusion), `scripts/` (dataset preparation and index building), `evaluation/` (queries, human labeling tools, metrics), `evaluation/experiments/` (labeled A/B evidence for the color experiment and the vanilla baseline), and `app.py` (Streamlit demo).

Conclusion. Vanilla CLIP remains a strong broad semantic baseline — measurably so on this corpus. But decomposing fashion queries and grounding garment constraints in category-aware region evidence improves top-k relevance precisely where global embeddings are weakest: fine-grained attribute queries and compositional multi-garment queries, lifting overall Precision@5 from 0.867 to 0.933 under identical human judgment. Scene-sparse contexts and explicit action/relation understanding remain open limitations, and the report documents where measured experimentation (color-contrast scoring) was deliberately not shipped because it lowered overall precision.